EE5373: Data Modeling Using R

Fall 2025

Lab 8: Final Project

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1. Introduction

The dataset analyzed is the **Hass Avocado Board Week Report 2025**, containing 5280 entries and 16 numeric and categorical features. Details regarding each column are listed below. The primary goal is to model and predict the variable `ASP.Current.Year`, representing average selling price, using relevant predictors derived from weekly avocado production and sales data across U.S. regions.

- **Report**: Metadata column ("Category"), identical across all rows.
- **Timeframe**: Always "Weekly". No variation;
- **Period**: Week number (e.g., 36, 35, ...).
- **Type**: Categorical variable indicating "Conventional" or "Organic".
- **Geography**: Region or city name where data were collected.
- **Current.Year.Week.Ending**: Week-ending date.
- **ASP.Current.Year**: Represnts the average selling price for the current week.
- **Total.Bulk.and.Bags**: Total units sold (bulk + bagged).
- **X4046**: small/medium Hass Avocados (3-5 oz).
- **X4225**: large Hass Avocados (8-10 oz);
- **X4770**: extra large Hass Avocados (10-15 oz)
- **TotalBagged**: Total number of bagged units sold.
- **Sml.Bagged**: Small bagged units.
- **Lrg.Bagged**: Large bagged units
- **X.Lrg.Bagged**: Extra-large bagged units;
- **Unknown.Bag.Size**: Units with unspecified bag size.

2. Data Cleaning and Preparation

The raw dataset was imported and cleaned by removing missing entries and redundant variables with no variation (`Report`, `Timeframe`, and `Type`). The cleaned dataset retained over 5280 rows and 16 relevant columns. Important categorical features were converted to factors as apporpriate.

```
raw_data <- read.csv("HAB-FlatFile-Weekly-2025-P1-P11.csv", header=
   TRUE)
dat <- na.omit(raw_data)
dat$Geography <- as.factor(dat$Geography)
```

Listing 1: Data loading and cleaning in R

Basic summary and correlation analysis were used to assess relationships between predictors, followed by visualization through scatterplot matrices and heatmaps to ensure data sanity.

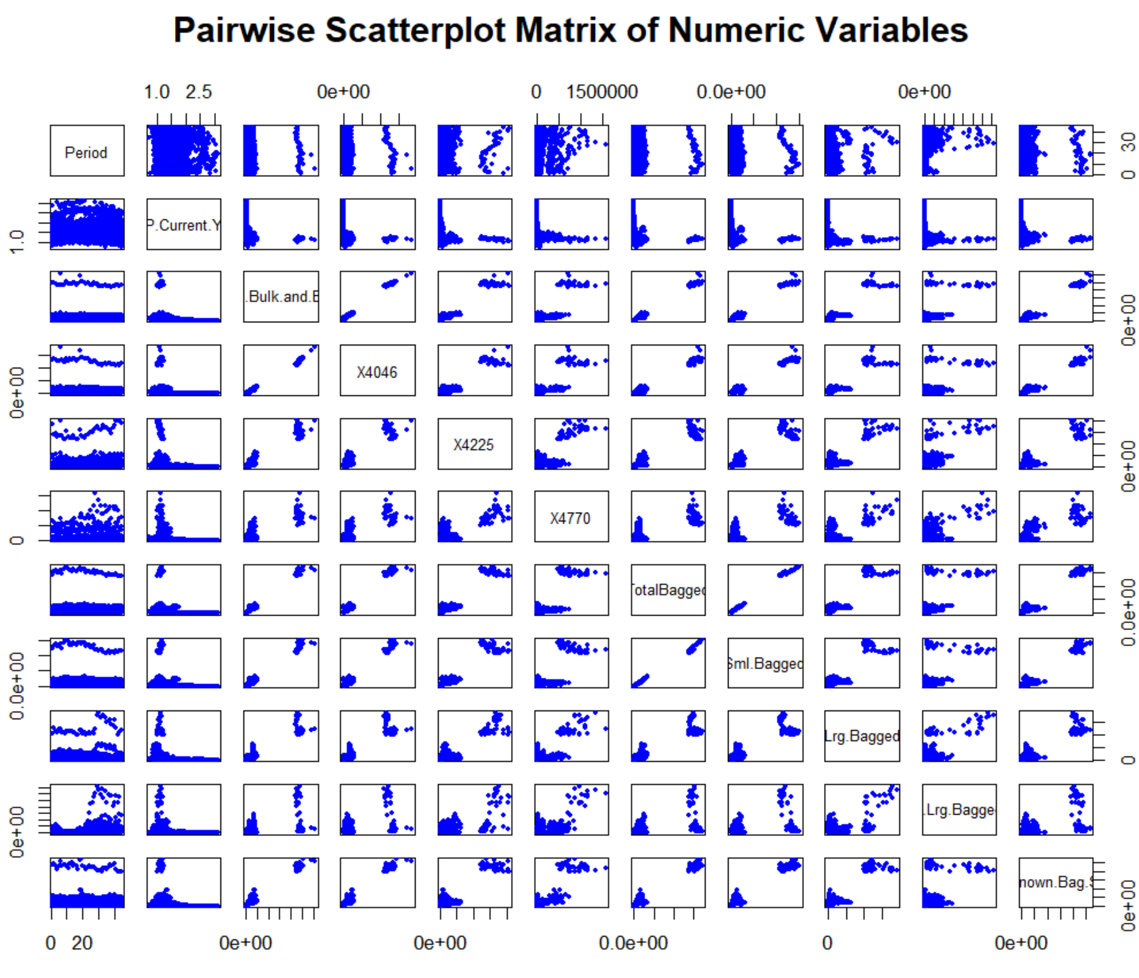

Figure 1: Pairwise scatterplot matrix of numeric variables.

3. Model Development and Feature Selection

The dataset was randomly split into 65% training and 35% testing subsets. A multiple linear regression model was developed to predict `ASP.Current.Year`. Backward elimination guided by statistical significance (p-values) was used to refine the model to six key predictors: `Period`, `Total.Bulk.and.Bags`, `X4046`, `X4225`, `Sml.Bagged`, and `Lrg.Bagged`.

```
set.seed(1234)
rows <- nrow(dat)
f <- 0.65
perm <- dat[sample(rows), ]
train.dat <- perm[1:floor(f * rows), ]
test.dat  <- perm[(floor(f * rows) + 1):rows, ]

model <- lm(ASP.Current.Year ~ Period + Total.Bulk.and.Bags + X4046
    + X4225
              + Sml.Bagged + Lrg.Bagged , data = train.dat)
summary(model)
```

Listing 2: General regression model after backward elimination.

4. Backward Elimination

In the process of backward elimination, columns `Report, Timeframe`, and `Type` were removed due to lack of variation. The variable `Current.Year.Week.Ending` was removed because its large number of unique categories introduced excessive complexity into the model. `Total.Bagged` was removed due to representing redundant information. `X.Lrg.Bagged` and `Unknown.Bag.Size` were removed due to their high p-values, indicating limited statistical significance. The effects of removing `Geography` are discussed below.

5. Initial Model Evaluation

To assess the impact of including `Geography` as a predictor, two linear regression models were evaluated: one with the `Geography` variable and one without it. Model performance was compared using RMSE and R^2 on both the training and testing sets.

Model Including `Geography`

- **Testing set:** RMSE = 0.2869, $R^2 = 0.345$
- **Training set:** RMSE = 0.429, $R^2 = 0.39$

Model Excluding `Geography`

- **Testing set:** RMSE = 0.3485, $R^2 = 0.03$
- **Training set:** RMSE = 0.367, $R^2 = 0.039$

These results clearly demonstrate that `Geography` is a critical predictor for average selling price. However, the variable includes a large number of categories, which complicates its use in a generalized linear model.

6. Hierarchical Modeling Solution

To address this challenge a **hierarchical modeling approach** was implemented. Separate regression models were built for each geographic region using the earlier parameter set identified in Listing 2.

Implementation

The hierarchical modeling was implemented in R through an automated iterative process:

1. Extract unique geographic regions from the training data
2. For each region, create training and testing subsets
3. Fit a linear regression model using region-specific data
4. Generate predictions and compute performance metrics
5. Store models and results for comparative analysis

Performance Summary

The hierarchical modeling approach yielded significantly improved predictive performance:

- **Average Test** R^2: 0.879 (substantial improvement over the global model's 0.345)
- **Average Test RMSE**: 0.09 (reduction of 67% from the global model's 0.287)
- **Performance Ranges**: $R^2 = [0.414, 0.968]$ and RMSE = [0.0394, 0.243]
- **Overfitting**: Minimal divergence from the unity line, (see Figure 2a and 2b, page 6) indicate consistent performance between training and testing datasets.

7. Conclusion

In summary, the global model could not accurately predict avocado prices because the `Geography` variable introduced too much complexity. By building separate models for each region, the predictions improved and became more consistent. This approach shows that avocado pricing patterns differ by location, and modeling each region individually provides a more reliable solution.

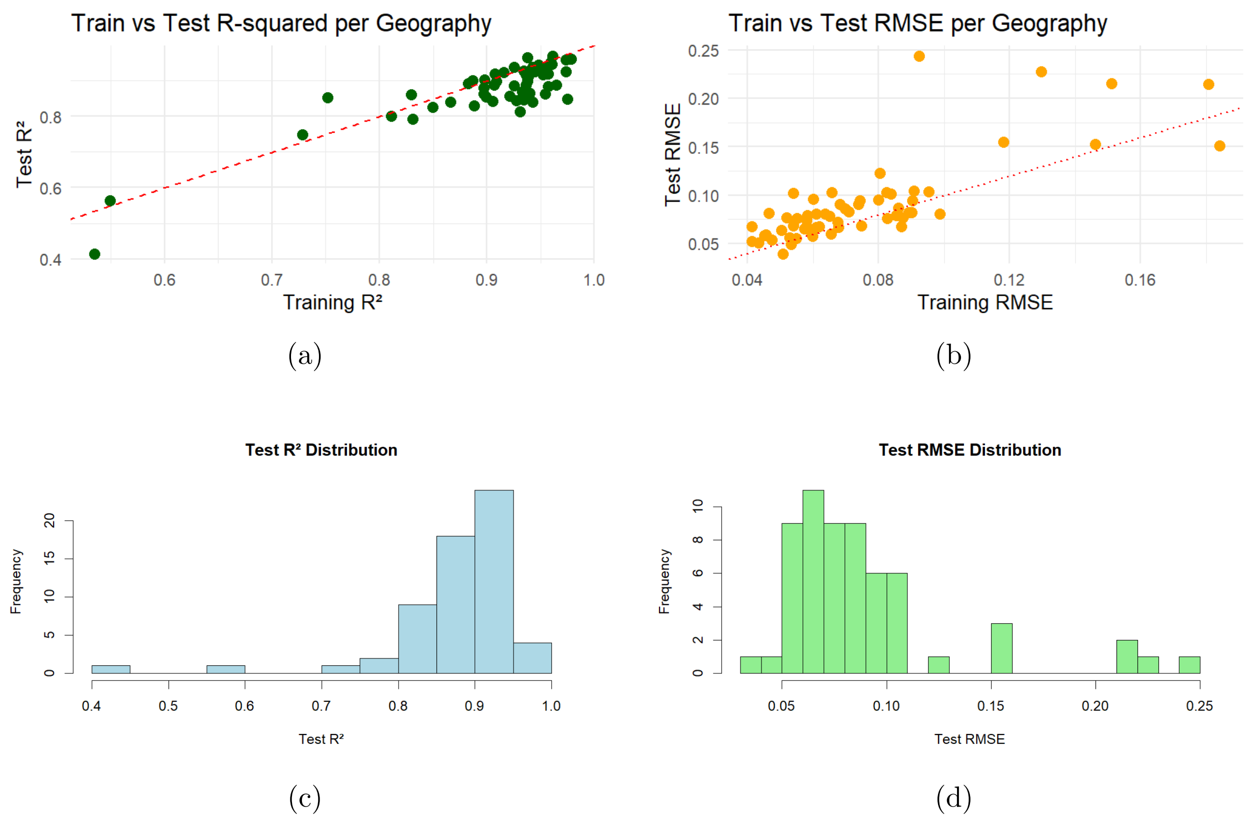

Figure 2: Model performance evaluation across geographical regions. (a, b) Comparison of training and testing metrics tests for overfitting. (c, d) Distribution of test metrics indicates overall prediction quality.